

Question

Oxalate is a good ligand and can form complexes with transition metals. Since oxalate solutions are strongly alkaline, metal ions that are prone to hydrolysis, such as Fe^{3+} , Al^{3+} , need to be converted to hydroxides first, then dissolved in oxalic acid, and neutralized with a base to obtain the corresponding oxalate complexes. This lengthy procedure is time-consuming, it is difficult to filter the hydroxides, the operation results in a lot of loss, and the yield is often not ideal.

Someone proposed the following experimental scheme to directly prepare the complex of oxalate with ferric iron: mixing 20 mL $3 \text{ mol} \cdot \text{L}^{-1}$ potassium oxalate hot solution and 8 mL $1.5 \text{ mol} \cdot \text{L}^{-1}$ ferric chloride solution, cooling to room temperature to obtain green needle-shaped crystals $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$, and measuring the volume of the solution at this time as 30 mL. Now we theoretically calculate the feasibility of this scheme.

Constants involved in this problem:

$\text{pK}_{\text{sp}}(\text{Fe}(\text{OH})_3) = 38.55$, $\lg K_{\text{f}}(\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}) = 20.2$, $\text{pK}_{\text{b1}}(\text{C}_2\text{O}_4^{2-}) = 9.73$. The saturated solubility of the product at room temperature is $4.9 \text{ g} \cdot 100 \text{ mL}^{-1}$, and the solubility of the two reactants is very high.

The secondary hydrolysis of oxalate is ignored in the calculation of this problem.

Which of the following options regarding this experimental scheme is correct (calculation error within 1% is considered correct):

- A.** When mixing potassium oxalate solution and ferric chloride solution, one can adopt the method of slowly and gradually adding the potassium oxalate solution to the ferric chloride solution, or the method of rapidly adding the ferric chloride solution to the potassium oxalate solution.
- B.** At room temperature, the expected yield of this experimental protocol is approximately 5.9 g.
- C.** Assuming iron hydroxide is not formed, the iron ion concentration is approximately 10^{-10} mol/L when the mixture is cooled to room temperature.

D. Assuming that iron hydroxide is not formed, when the mixture is cooled to room temperature, the concentration of hydrogen oxalate ion is approximately 10^{-10} mol/L.

E. The mass of iron-containing species precipitate, excluding the product, generated by this experimental protocol is approximately 0.3 mg.

F. This experimental protocol is completely infeasible.

G. All of the above options are incorrect.

Answer

Correct Answer: **G**

Detailed Explanation

In this experiment, it is necessary to ensure that ferric ions do not hydrolyze to form ferric hydroxide, which would affect the yield of potassium ferrioxalate; therefore, it is necessary to ensure that oxalate is in excess or locally in excess to avoid local overconcentration of ferric ions and hydrolysis; therefore, the ferric solution needs to be slowly dripped into the oxalate solution, or the oxalate solution needs to be quickly dripped into the ferric solution, so option A is incorrect.

CHECKPOINT

1 PTS

It is necessary to ensure that oxalate is in excess or locally in excess to avoid local overconcentration of ferric ions and hydrolysis

Obviously, the reaction ratio of oxalate to ferric iron is 3:1, thus oxalate is in excess; at this time, calculate the theoretical yield based on the amount of iron. Assuming all the iron is converted into the product, the mass of the product is $0.008 \times 1.5 \times 491.1 = 5.89$ g. However, the dissolution loss needs to be considered. Considering the theoretical yield after dissolution is $5.89 - 4.9/100 \times 30 = 4.42$ g, so option B is incorrect.

CHECKPOINT

1 PTS

The theoretical yield after considering dissolution loss is 4.42 g

Since the coordination stability constant of potassium ferrioxalate is relatively large, it can be approximated that most of the iron ions exist in the form of complexes;

CHECKPOINT

1 PTS

It can be approximated that most of the iron ions exist in the form of complexes

Since the secondary hydrolysis of oxalate can be ignored, the remaining concentration after deducting the part forming the complex is:

$$[\text{C}_2\text{O}_4^{2-}] = (20 \times 3 - 3 \times 8 \times 1.5)/30 = 0.8 \text{ mol/L}$$

CHECKPOINT

1 PTS

$$[\text{C}_2\text{O}_4^{2-}] = 0.8 \text{ mol/L}$$

Calculate the saturation concentration through the solubility of the product in water:

$$[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}] = 4.9/491.1/0.1 = 0.1 \text{ mol/L}$$

CHECKPOINT

1 PTS

$$[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}] = 0.1 \text{ mol/L}$$

Then calculate the concentration of ferric ions through the stability constant of the complex:

$[\text{Fe}^{3+}] = [\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}]/(\text{K}_f \cdot [\text{C}_2\text{O}_4^{2-}]^3) = 0.1/10^{20.2} \cdot 0.8^3 = 10^{-20.9} \text{ mol/L}$, the concentration is extremely small, which is approximately reasonable.

CHECKPOINT

1 PTS

$[\text{Fe}^{3+}] = 10^{-20.9} \text{ mol/L}$, approximately reasonable

Ignoring the secondary hydrolysis of oxalate, we have $[\text{HC}_2\text{O}_4^-] = [\text{OH}^-] = \sqrt{[\text{C}_2\text{O}_4^{2-}]\text{K}_{\text{b1}}} = 10^{-4.91} \text{ mol/L}$

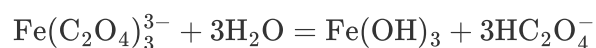
CHECKPOINT

1 PTS

$[\text{HC}_2\text{O}_4^-] = [\text{OH}^-] = 10^{-4.91} \text{ mol/L}$

Therefore, options C and D are both incorrect.

The hydrolysis equation of ferrioxalate ion is



The equilibrium constant of the reaction is $K = K_{\text{sp}}K_{\text{b1}}^3/K_{\text{f}} = 10^{-10.84} > Q = [\text{HC}_2\text{O}_4^-]^3/[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}] = 10^{-13.7}$, so ferric hydroxide will be generated;

CHECKPOINT

1 PTS

The hydrolysis equilibrium constant of ferrioxalate ion is $K = 10^{-10.84} > Q = 10^{-13.7}$, so ferric hydroxide will be generated

According to the equilibrium constant calculation, $[\text{HC}_2\text{O}_4^-]' = (K[\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}])^{1/3} = 10^{-3.95} \text{ mol/L}$ at hydrolysis equilibrium

CHECKPOINT

1 PTS

$[\text{HC}_2\text{O}_4^-]' = 10^{-3.95} \text{ mol/L}$ at hydrolysis equilibrium

The amount of precipitate is $n(\text{Fe}(\text{OH})_3) = ([\text{HC}_2\text{O}_4^-]' - [\text{HC}_2\text{O}_4^-]) \times 0.030/3 = 10^{-6} \text{ mol}$, which is calculated as 0.11 mg; option E is incorrect.

CHECKPOINT

1 PTS

The mass of ferric hydroxide generated is 0.11 mg

Since the mass of ferric hydroxide generated is much smaller than the mass of the product, it can be considered that the experimental scheme is feasible, so option F is incorrect.

CHECKPOINT

1 PTS

The mass of ferric hydroxide generated is much smaller than the mass of the product, so it can be considered that the experimental scheme is feasible

In summary, options A-F are all incorrect, and option G is correct.